=== UNIT 6: FOURIER SERIES AND HARMONIC ANALYSIS ===

>> 6.1 Periodic Functions and Dirichlet’s Conditions

Fourier analysis decomposes arbitrary periodic engineering signals into linear combinations of pure sinusoidal harmonics.
* Periodic Functions: A function f(x) is periodic with period T if f(x + T) = f(x) for all x in its domain.
* Dirichlet’s Conditions for Fourier Convergence: Any function f(x) defined in interval (-c, c) can be expanded into a convergent Fourier series provided:
1. f(x) is periodic, single-valued, and bounded.
2. f(x) has at most a finite number of jump discontinuities in any single period.
3. f(x) has at most a finite number of local maxima and minima in any single period.
* Convergence at Discontinuities: At a point of finite discontinuity x0, the Fourier series converges to the arithmetic mean of its left- and right-hand limits: f(x0) = [f(x0-) + f(x0+)] / 2.

>> 6.2 Euler’s Formulae for Fourier Coefficients

* Fourier Series Representation in Interval (-pi, pi):
f(x) = a0 / 2 + sum_{n=1}^inf [an * cos(nx) + bn * sin(nx)].
* Euler’s Formulae:
a0 = (1 / pi) * int_{-pi}^{pi} f(x) dx.
an = (1 / pi) * int_{-pi}^{pi} f(x) * cos(nx) dx (for n = 1, 2, 3, ...).
bn = (1 / pi) * int_{-pi}^{pi} f(x) * sin(nx) dx (for n = 1, 2, 3, ...).
* Fourier Series for Arbitrary Period T = 2L (Interval -L to L):
f(x) = a0 / 2 + sum_{n=1}^inf [an * cos(n*pi*x / L) + bn * sin(n*pi*x / L)], where:
a0 = (1 / L) * int_{-L}^L f(x) dx.
an = (1 / L) * int_{-L}^L f(x) * cos(n*pi*x / L) dx.
bn = (1 / L) * int_{-L}^L f(x) * sin(n*pi*x / L) dx.

>> 6.3 Even and Odd Functions (Half-Range Expansions)

* Even Functions (f(-x) = f(x), symmetric about y-axis):
Product f(x)*sin(nx) is odd => bn = 0 (No sine terms).
Fourier Cosine Series: a0 = (2 / L) * int_0^L f(x) dx; an = (2 / L) * int_0^L f(x)*cos(n*pi*x / L) dx.
* Odd Functions (f(-x) = -f(x), symmetric about origin):
f(x) and f(x)*cos(nx) are odd => a0 = 0 and an = 0 (No cosine terms).
Fourier Sine Series: bn = (2 / L) * int_0^L f(x)*sin(n*pi*x / L) dx.
* Half-Range Series: A function defined only on interval (0, L) can be expanded into either:

1. Half-Range Sine Series: Extend f(x) as an odd function into (-L, 0). Contains only sine terms.
2. Half-Range Cosine Series: Extend f(x) as an even function into (-L, 0). Contains only cosine terms.

[SOLVED PROBLEM 1: Fourier Series of Square Wave and Gregory-Leibniz Formula]

Problem: Obtain the Fourier series for the square wave function defined by f(x) = -k for -pi < x < 0, and f(x) = +k for 0 < x < pi, with f(x + 2*pi) = f(x). Hence deduce the Gregory-Leibniz series: 1 - 1/3 + 1/5 - 1/7 + ... = pi / 4.

Solution Step 1: Symmetry inspection: f(-x) = -f(x). Thus f(x) is an ODD function.

Consequently: a0 = 0 and an = 0 for all n.

Solution Step 2: Compute Fourier sine coefficients bn:

bn = (2 / pi) * int_0^pi f(x) * sin(nx) dx = (2 / pi) * int_0^pi k * sin(nx) dx

bn = (2*k / pi) * [-cos(nx) / n]_0^pi = (2*k / (n*pi)) * [1 - cos(n*pi)].

Since cos(n*pi) = (-1)^n:

- If n is EVEN: 1 - (-1)^n = 1 - 1 = 0 => bn = 0.
- If n is ODD: 1 - (-1)^n = 1 - (-1) = 2 => bn = 4*k / (n*pi).

Solution Step 3: Write out Fourier Series:

f(x) = (4*k / pi) * [sin(x) + sin(3x)/3 + sin(5x)/5 + sin(7x)/7 + ...].

Solution Step 4: Deduce numerical sum at x = pi/2:

At x = pi/2, f(pi/2) = k.

sin(pi/2) = 1, sin(3*pi/2) = -1, sin(5*pi/2) = 1, sin(7*pi/2) = -1.

Substitute: k = (4*k / pi) * [1 - 1/3 + 1/5 - 1/7 + ...].

Divide by k and multiply by pi/4:

1 - 1/3 + 1/5 - 1/7 + ... = pi / 4. (Hence Proved).

[SOLVED PROBLEM 2: Fourier Expansion of Parabolic Function f(x) = x^2]

Problem: Find the Fourier series of f(x) = x^2 in the interval -pi <= x <= pi. Hence prove that:

1/1^2 - 1/2^2 + 1/3^2 - 1/4^2 + ... = pi^2 / 12, and 1/1^2 + 1/2^2 + 1/3^2 + ... = pi^2 / 6.

Solution Step 1: Function parity: f(-x) = (-x)^2 = x^2 = f(x). f(x) is EVEN.

Therefore: bn = 0 for all n.

Solution Step 2: Calculate a0:

a0 = (2 / pi) * int_0^pi x^2 dx = (2 / pi) * [x^3 / 3]_0^pi = (2 / pi) * (pi^3 / 3) = 2*pi^2 / 3.

Constant term: a0 / 2 = pi^2 / 3.

Solution Step 3: Calculate an using integration by parts:

an = (2 / pi) * int_0^pi x^2 * cos(nx) dx.

Let u = x^2, dv = cos(nx) dx => v = sin(nx)/n, v1 = -cos(nx)/n^2, v2 = -sin(nx)/n^3.

Using generalized parts: int u*v dx = u*v - u'*v1 + u''*v2
= [x^2 * (sin(nx)/n) - (2x)*(-cos(nx)/n^2) + (2)*(-sin(nx)/n^3)]_0^pi.
Evaluating at 0 and pi: sin(n*pi) = 0, so only cosine term survives:
an = (2 / pi) * [2*pi * cos(n*pi) / n^2] = (4 / n^2) * (-1)^n.
Solution Step 4: Write Fourier Series:
x^2 = pi^2 / 3 + 4 * sum_{n=1}^inf [((-1)^n / n^2) * cos(nx)]
x^2 = pi^2 / 3 - 4 * [cos(x)/1^2 - cos(2x)/2^2 + cos(3x)/3^2 - cos(4x)/4^2 + ...].
Solution Step 5: Deduction 1 (Set x = 0):
0 = pi^2 / 3 - 4 * [1/1^2 - 1/2^2 + 1/3^2 - 1/4^2 + ...]
=> 4 * [1/1^2 - 1/2^2 + 1/3^2 - 1/4^2 + ...] = pi^2 / 3
=> 1/1^2 - 1/2^2 + 1/3^2 - 1/4^2 + ... = pi^2 / 12. (Proven).
Solution Step 6: Deduction 2 (Set x = pi):
pi^2 = pi^2 / 3 - 4 * [-1/1^2 - 1/2^2 - 1/3^2 - ...] = pi^2 / 3 + 4 * [1/1^2 + 1/2^2
+ 1/3^2 + ...]
pi^2 - pi^2 / 3 = 2*pi^2 / 3 = 4 * sum_{n=1}^inf (1 / n^2)
=> sum_{n=1}^inf (1 / n^2) = (2*pi^2 / 3) / 4 = pi^2 / 6. (Basel Problem Proven).

[SOLVED PROBLEM 3: Half-Range Sine Series for Linear Ramp]

Problem: Find the half-range Fourier sine series for f(x) = x in the interval 0 < x < 2
(L = 2).
Solution Step 1: For half-range sine series, a0 = 0 and an = 0.
Solution Step 2: Compute bn with L = 2:
bn = (2 / L) * int_0^L f(x) * sin(n*pi*x / L) dx = (2 / 2) * int_0^2 x * sin(n*pi*x / 2)
dx.
bn = int_0^2 x * sin(n*pi*x / 2) dx.
Solution Step 3: Integrate by parts (u = x, dv = sin(n*pi*x / 2) dx):
v = - [2 / (n*pi)] * cos(n*pi*x / 2).
bn = [- x * (2 / (n*pi)) * cos(n*pi*x / 2)]_0^2 - int_0^2 [- (2 / (n*pi)) *
cos(n*pi*x / 2)] dx
bn = [- 2 * (2 / (n*pi)) * cos(n*pi) - 0] + [4 / (n^2*pi^2) * sin(n*pi*x / 2)]_0^2.
Since sin(n*pi) = 0 and sin(0) = 0, the second bracket is zero.
bn = - (4 / (n*pi)) * (-1)^n = (4 / (n*pi)) * (-1)^{n+1}.
Solution Step 4: Write series expansion:
f(x) = sum_{n=1}^inf [(4 / (n*pi)) * (-1)^{n+1} * sin(n*pi*x / 2)]
f(x) = (4 / pi) * [sin(pi*x/2) - (1/2)*sin(2*pi*x/2) + (1/3)*sin(3*pi*x/2) - ...].

[SOLVED PROBLEM 4: Half-Range Cosine Series for f(x) = x*(pi - x)]

Problem: Obtain the half-range cosine series for f(x) = pi*x - x^2 in the range 0 <= x
<= pi.
Solution Step 1: For half-range cosine series in (0, pi), L = pi. bn = 0.

Solution Step 2: Compute a0:
a0 = (2 / pi) * int_0^pi (pi*x - x^2) dx = (2 / pi) * [pi*x^2/2 - x^3/3]_0^pi
a0 = (2 / pi) * [pi^3/2 - pi^3/3] = (2 / pi) * (pi^3 / 6) = pi^2 / 3.
Constant term: a0 / 2 = pi^2 / 6.
Solution Step 3: Compute an using integration by parts:
an = (2 / pi) * int_0^pi (pi*x - x^2) * cos(nx) dx.
Using generalized formula (u = pi*x - x^2, u' = pi - 2x, u'' = -2):
an = (2 / pi) * [(pi*x - x^2)*(sin(nx)/n) - (pi - 2x)*(-cos(nx)/n^2) +
(-2)*(-sin(nx)/n^3)]_0^pi.
Evaluating at 0 and pi (sin terms vanish):
At x = pi: - (pi - 2*pi)*(-cos(n*pi)/n^2) = - (-pi)*(-(-1)^n / n^2) = - pi*(-1)^n / n^2.
At x = 0: - (pi - 0)*(-cos(0)/n^2) = - pi*(-1/n^2) = + pi / n^2.
an = (2 / pi) * [- pi*(-1)^n / n^2 - pi / n^2] = - (2 / n^2) * [1 + (-1)^n].
- If n is ODD: 1 + (-1)^n = 1 - 1 = 0 => an = 0.
- If n is EVEN: 1 + (-1)^n = 2 => an = - 4 / n^2. Let n = 2m: a_{2m} = - 4 /
(2m)^2 = - 1 / m^2.
Solution Step 4: Write Half-Range Cosine Series:
pi*x - x^2 = pi^2 / 6 - sum_{m=1}^inf [(1 / m^2) * cos(2mx)]
pi*x - x^2 = pi^2 / 6 - [cos(2x)/1^2 + cos(4x)/2^2 + cos(6x)/3^2 + ...].

>> 6.4 University Review & Examination Questions

* Q1: State Dirichlet's conditions for the existence of a Fourier series of a function f(x) in (-pi, pi).
Answer: Dirichlet's conditions guarantee that a function f(x) defined on (-pi, pi) has a convergent Fourier series: (1) f(x) must be single-valued and bounded everywhere in (-pi, pi). (2) f(x) must have at most a finite number of jump discontinuities within any single period. (3) f(x) must have at most a finite number of local maxima and local minima in that interval. When these conditions hold, the series converges to f(x) at every point of continuity, and converges to [f(x+) + f(x-)] / 2 at any point of discontinuity.
* Q2: Explain Parseval's identity for Fourier series and its application in summing infinite numerical series.
Answer: Parseval's identity equates the mean-square value of a periodic function to the sum of the squares of its Fourier coefficients: (1 / 2L) * int_{-L}^L [f(x)]^2 dx = (a0 / 2)^2 + (1 / 2) * sum_{n=1}^inf (an^2 + bn^2). In the interval (-pi, pi), this becomes: (1 / pi) * int_{-pi}^{pi} [f(x)]^2 dx = a0^2 / 2 + sum_{n=1}^inf (an^2 + bn^2).
Parseval's theorem is extraordinarily powerful in computing higher-order infinite numerical series (such as sum 1/n^4 or sum 1/n^6) by simply evaluating the definite integral of [f(x)]^2.

>> 6.5 Key Takeaways and Revision Points

* Euler's formulas determine harmonic coefficients a0, an, and bn.

* Even functions yield pure cosine series (bn = 0); odd functions yield pure sine series (a0 = an = 0).
* At points of jump discontinuity, Fourier series converges to the midpoint [f(x+) + f(x-)] / 2.
* Half-range series extend functions defined on (0, L) into full even or odd periodic domains.